

# Open Science Indicators Methods documentation for v10 Public Data

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Public Library of Science (2022) PLOS Open Science Indicators. Figshare. Dataset (version 10). <https://doi.org/10.6084/m9.figshare.21687686>.

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## Assembly of PLOS-Dataset\_v10\_Jul25.csv:

The version 9 PLOS dataset (PLOS-Dataset\_v9\_Dec24.csv) was combined with new data gathered following the same methodology as version 1 with a small deviation in how the articles were downloaded. For v10, articles were collected using the source code shared via source code located at <https://github.com/PLOS/allofplos>, rather than the python library as was used in previous versions. This change meant we gathered an additional 6,449 articles published between 1/1/18 and 30/9/24 that were missing from previous versions of the dataset, including 59 articles from two new PLOS journals (*PLOS Mental Health* and *PLOS Complex Systems*) which were published from June 2024 onwards. In addition, we selected a set of 9980 additional articles with a publication date between 01/10/2024 and 30/3/2025 for the v10 dataset. We included articles designated as research articles (article type was “Research Article”, “Meta-Research Article”, or “Pre-Registered Research Article”).

In addition to these criteria we only included articles with a Data Availability Statement identified within the XML file, and at least one of the following sections in the XML: materials|methods, and supplementary material. Inclusion of articles with all three section tags was prioritized (i.e. Data Availability Statement, materials|method, and supplementary material).

However, articles missing one of the non-mandatory text section tags (i.e. missing materials|method or supplementary material) are included using a full-text analysis to ensure any information provided in an unlabeled section was included in the analysis. The v10 dataset has a total of 138,995 PLOS articles. All articles were analysed using the same algorithm as used in v5 and later versions. Author and country information was gathered following the method first used in v9.

### Assembly of Comparator-Dataset\_v10\_Jul25.csv:

The version 9 Comparator dataset (Comparator-Dataset\_v9\_Dec24.csv) was combined with new data gathered following the same methodology as version 9. An additional comparator set of 1,971 Open Access articles published in non-PLOS journals between 01/10/24 and 30/3/2025 was assembled for v10. An additional 844 articles published between 1/1/18 and 30/9/24 were also to mirror the additional articles from this time period added to the PLOS corpus.

The article selection method used for versions 1 to 6 was as follows: To ensure a broad subject area match between the PLOS dataset and the comparators, we downloaded the major MeSH terms from PubMed Central (PMC) for the 61,318 PLOS articles (v1 dataset). We obtained a list of 11,728 major MeSH terms that appear between 1 and 1083 times in the corpus. Terms that appear on many PLOS articles (e.g. COVID19) correspondingly appear many times in this list. We then randomly selected a 1200-term subset with replacement, such that selected terms can appear multiple times in the created list if they appear frequently in the MeSH distribution. The same list of MeSH terms was used to sample the additional comparator articles for v6. For the new articles included in v7 - articles published between 1/1/24 and 31/3/24, a new set of MeSH terms was used. These were obtained by downloading the major MeSH terms from PLOS articles published between 1/1/22 and 31/12/23. These were used to sample the PMC articles for the comparator set. From this we obtained a list of 10,351 major MeSH terms.

For versions 1 to 6, articles were chosen for the comparator datasets as follows. We searched within PubMed Central's Open Access corpus (<https://www.ncbi.nlm.nih.gov/pmc/tools/openftlist/>) for each MeSH term. For each candidate article matching the term, we excluded articles whose XML indicated that the publisher was PLOS, and constrained the results to articles of type "Journal Article" published in the same time period; articles already in the comparator dataset were also excluded. A random article was chosen per query term and added to the comparator dataset.

For version 8, we realized that the sampling of new articles for v7 (i.e. those published between 1/1/24 and 31/3/24) was biased towards articles published in January 2024 due to changes in the PMC database. Specifically, the number of articles returned at the time of sampling for a search for any MeSH term on PMC fell quickly to zero after January 2024. For v8, we resampled the articles published in this period by instead searching with the relevant MeSH terms on PubMed (as opposed to PubMedCentral) and constraining the search to articles with full text available on PMC. The comparator set was re-collected in this way for articles published between 1/1/2024 and 31/3/2024. The additional comparator articles for versions 9 onwards were collected by searching on PubMed.

OpenAlex information on author country and topics was added to the comparator dataset as outlined in the assembly of the PLOS dataset.

The comparator dataset was then processed with methodology analogous to the PLOS dataset detailed above using the nxml files downloaded from PMC. Due to differences in the provided metadata between PMC nxml and PLOS XML, the metadata collection methods differ between the two corpora. For example, a lack of consistent availability of certain metadata in the PMC nxml files, not all metadata fields were provided for each article. An additional field is included in the comparator set to provide further context when interpreting the results: in place of listed disciplines found in the PLOS XML, the list of assigned major MeSH terms is included for each article.

## Preprint Detection:

For version 10 a new preprint detection method was employed and run across the whole dataset. First we gather Crossref information about the published articles of the main (PLOS) corpus and comparator sets. Then we enrich records that are missing the abstract using OpenAlex. We filter the corpus to the records listed as preprints in Crossref (<https://www.crossref.org/learning/public-data-file/>) and DataCite (<https://datafiles.datacite.org/datafiles/public-2024>) as the matching pool, including DataCite because it indexes ArXiv.

This results in a global list of ~4 million preprints. We then create embeddings—numerical vector representations—for each record by tokenizing the title and abstract and passing these through a transformer model called SPECTER2 ([https://huggingface.co/allenai/specter2\\_base](https://huggingface.co/allenai/specter2_base)). In this embedding space, the distance between different records corresponds to their semantic and scientific similarity. By embedding the focal articles in the same space, we can produce preprint candidate lists based on which preprints are near the focal article, ordering them based on distance/similarity.

We then re-order the candidate list by passing it to a statistical model that does fuzzy matching on authors' family names between the published article and the candidates. The final ranking score is a linear combination of the author similarity and the embedding cosine similarity, with weights that we have chosen by hand. Using this score, we match each published article with the highest-scoring preprint. Importantly, at this stage, we produce a match for every published article, whether or not it has a preprint. False matches are filtered in the next step. Based on our testing, we believe that, for articles that do have a preprint, the correct preprint appears as the top-ranked candidate 96% of the time. In other words, at this stage, we achieve a Recall@1 of 96%.

Finally, we verify all of the matches using a language model, which also filters out the negative matches. This model is a fine-tuned version of Qwen/Qwen3-4B using low-rank adaptation (rank 8) trained with reinforcement learning with verifiable reward (RLVR) for two epochs, a total of 2250 gradient steps, on 17M tokens. The training dataset was assembled by annotators at DataSeer and contains positive matches and hard negatives. At inference time, the model produces a chain-of-thought and a prediction about whether the article-preprint pair is a match or not. We sample six predictions and rerank them using Qwen/Qwen3-32B instructed to score each of the completions. The highest scoring completion is chosen as the model's

prediction for the pair. The effect is to filter the list of preprint-article pairs and add a reasoning trace to each one.

For Crossref we mainly look for articles of subtype="Preprint". We do not use "posted-content" as this includes non-preprint information. We also match "group-title" to a list of known preprint servers, finding that this step adds a number of preprints that were not listed as preprints in Crossref. We apply the same matching to the "publisher" field, finding that the preprint server is occasionally listed there. For articles explicitly listed as preprints, we also verify the DOI and remove records if neither the publisher nor group-title contain a preprint server.

For DataCite we only verify that the record's server (called "publisher" in DataCite cataloging) matches our list of known preprint servers including arXiv, bioRxiv, and PsyArXiv, as well as a number of other (non-"Xiv") preprint servers

## Data and Code Generation:

We first determined if each article had generated one or more datasets to allow consideration of OSIs as both a percentage of all articles as well as for only articles that had shareable datasets, as desired. To do this, we applied a custom Natural Language Processing (NLP) model (<https://github.com/DataSeer/dataseer-ml>) to the Methods section of the article to detect sentences describing data collection. When the article did not have a detectable Methods section, the full text of the article was analyzed. The model also detects sentences describing the re-use of existing datasets. Since re-analysis of existing datasets frequently requires additional manipulation of the data – and hence the creation of a new shareable dataset – we counted re-use of existing data as 'data generation'.

We detected the generation of shareable code objects with a similar protocol. Sentences in the Methods text of each article were processed by a NLP model designed to detect keywords associated with code generation or script use (e.g. 'script'). An article was also designated as 'generating code' if it mentioned command line software (e.g. Mathematica) or commonly used coding environments (e.g. R or Python).

## Data and Code Sharing:

We then assessed whether data were shared within the supplementary files of the article or on an online repository. To determine whether datasets were shared as supplementary files we first excluded image files, specifically files with the mime\_type=image or the type .jpg, .tif, .png. We then determined if the file contained data by applying a NLP model to the caption, title, and file type. In addition to this, we used a similar NLP model to analyze sentences from the text in sections where data sharing is usually described (ie. Methods, and Data Availability Statements) to determine if an article shared data on a repository.

We applied a similar workflow to determine whether articles shared code, either as supplemental material or on a public repository. To complement this assessment we also provide DOIs and URLs mentioned in text that are likely to be involved with the code or data sharing. These are taken from text sections that describe sharing and are provided as a complete list of resources shared in the article. We identify commonly used repositories where possible from these URLs and DOIs (see OSI-Repository-List\_v1\_Dec22.xlsx). We used

domain knowledge and frequency of URL domain to identify commonly used online resources; we then verified repositories that hosted code and data before adding them to the detected repository list. This list is not a complete record of every repository used in this dataset, and will continue to be built upon with future data releases. A more inclusive assessment of data sharing was captured in the “Data\_location” column, which assigns data as being shared online, in supplementary material or both. The “online” category includes repositories as well as other online locations, such as lab websites. It, therefore, includes a greater number of articles although the majority of those sharing “online” are doing so via a repository.

## Protocol detection

We identified and tracked protocol sharing in research articles by analyzing JATS XML files from PLOS and PMC, enriched with metadata from Crossref and OpenAlex and standardized using the NCBI ID Converter. Protocols were sourced from 12 identified platforms, including repositories and journals, and filtered to exclude non-protocol content using Crossref metadata and DOI patterns. Protocol URLs were normalized and supplemented using OpenAlex’s location field. We then examined three types of sharing events—references, in-text links, and supplemental information (SI) files—matching DOIs and URLs against a cleaned allowlist of verified protocols. SI files were further filtered using a denylist to exclude irrelevant documents. This process yielded a structured dataset of protocol-sharing events across a curated corpus of articles. Further details are given in [Protocols-Methods-Statement\\_Sep23.pdf](#), which outlines the methods associated with the preliminary release of the protocols indicator.

## Study registration

We identified study registration events in research articles by developing a corpus-aware detection pipeline grounded in regular expression matching. Registrations were sourced from 31 study registries spanning clinical trials, systematic reviews, animal studies, and general-purpose platforms such as OSF and AsPredicted. We designed tailored regular expressions to capture registry names, abbreviations, PIDs, DOIs, and URLs as they appear in article text, links, and supplementary materials. Using XML versions of articles (primarily in JATS format), we extracted and classified content across section types to locate registration events and their context. The pipeline distinguishes between incidental mentions and true registrations by analyzing the proximity of registry identifiers and context-specific language. For OSF registrations, we matched against a comprehensive list of known registration IDs and DOIs. Reference lists were not analyzed due to low reporting rates, and standalone supplementary files were excluded from processing. The resulting tool achieved high performance (F1 score = 0.94) in detecting study registrations across both PLOS and PMC articles. Further details are given in [Study-Registration-Methods-Statement\\_Jun24.pdf](#), which outlines the methods associated with the preliminary release of the study registrations indicator.

# Open Science Indicators Accuracy rates for preprints v10 release

As with previous versions we have aimed for a minimum accuracy rate of at least 85%. We took 300 (total) publication DOIs from both PLOS and the comparator set for the ground truth sample. The Curators were given "candidate" preprint DOIs for about two thirds of the ground truth sample and if the candidate was not determined to be a preprint, the Curators then searched using Google the publication title along with "preprint" and also "xiv". The Curators were quite strict in their determination if a preprint matched a publication. They looked for the obvious features, like overlapping title/authors, and also for things like the same statistics/dates/locations reported in the abstracts and whether the date of the preprint was before the publication.

*Agreement between manual annotation of publications for preprints and the output of the algorithm for a sample of 300 articles.*

|                    |     | Manual annotation |     |     |
|--------------------|-----|-------------------|-----|-----|
|                    |     | Yes               | No  |     |
| DataSeer algorithm | Yes | 85                | 4   | 89  |
|                    | No  | 5                 | 206 | 211 |
|                    |     | 90                | 210 |     |

*Accuracy metrics for the sample of 300 articles used in the ground truth sample to assess the performance of the new preprint detection method.*

|             |      |
|-------------|------|
| Accuracy    | 0.97 |
| F1          | 0.95 |
| Precision   | 0.96 |
| Sensitivity | 0.94 |
| Specificity | 0.98 |

## Open Science Indicators Accuracy rates for v5 release

We have aimed for a minimum accuracy rate of at least 85% for all indicators and content sources. The accuracy rate is calculated by randomly selecting 100-200 articles from each corpus and checking them by hand to identify false positives and false negatives. These measures are then used to calculate the overall accuracy of the DataSeer assignments. For PLOS articles, all indicators meet our goal accuracy level but for the comparator corpus data sharing accuracy rates are below this minimum.

*Indicator accuracy rates reported by DataSeer.*

| Indicator        | Accuracy assessment<br>PLOS articles | Accuracy assessment<br>Non-PLOS articles |
|------------------|--------------------------------------|------------------------------------------|
| Data generation  | 88%                                  | 89%                                      |
| Data sharing     | 85%                                  | 81%                                      |
| Code generation  | 85%                                  | 92%                                      |
| Code sharing     | 97%                                  | 94%                                      |
| Preprint sharing | 94%                                  | 96%                                      |

## Open Science Indicators accuracy rates for v1 release

Allegra Pearce (updated 10-01-2023)

The accuracy rates presented are for the v1 dataset: Public Library of Science (2022) PLOS Open Science Indicators. Figshare. Dataset (version 1).

<https://doi.org/10.6084/m9.figshare.21687686>.

Below are the calculated accuracy results for the DataSeer analysis and ODDPub (Riedel et al., 2020) for both data and code. We plan to share detailed accuracy results for other indicators, e.g. preprints, in the future. For both the PLOS and Comparator corpus, results are calculated for a 100 article ground truth set manually curated by DataSeer. The manual coding for the accuracy estimates is based on a full human read-through of the article plus testing of the web links. Data Generation is determined by the presence of one or more data related sentences, either for the generation of new data or the re-use of existing datasets. In each set we've provided accuracy rates, sensitivity, specificity, precision, and F-scores. In addition to this we have provided confusion matrices with the true and false positive and negative labels for each metric (per dataset), these values are what the accuracy measures are based on. Below is

a brief definition of each of the accuracy measures.

*Accuracy rate (%)*: proportion of correctly labeled articles

*Recall/Sensitivity*: ratio of correctly labeled positive cases to total true positive cases

*Specificity*: ratio of correctly labeled negative cases to total true negative cases

*Precision*: ratio of correctly labeled positive cases to all cases labeled positive

*F-score*: harmonized mean of precision and recall (also called sensitivity)

Each of these specialized metrics shows a particular piece of information and is very helpful in diagnosing and directing continual improvements in development.

ODDPub's published F-scores are 0.73 for open data and 0.64 for open code. As a note, the authors also indicate [in their publication](#) that the open code assessment (F-score) is likely inaccurate due to the very low occurrence rates of code sharing (11 out of 792, Riedel et al., 2020). The effects of low occurrence rates are also apparent in the PLOS and PMC Comparator corpora studied here

Open science indicators with unbalanced cases (i.e. have many more positive or negative cases than the opposite) can show different impacts per correct or incorrect label in each accuracy metric. Metrics like sensitivity and specificity are a proportion and are sensitive to the total number of true cases. A single incorrect label can have a much larger impact on a proportion when there are fewer cases than when there are many, and as a result a single incorrect/correct label can have a much larger impact on an accuracy metric, while having a much smaller impact on overall accuracy in unbalanced datasets where there are fewer total true cases.

As an example, in data generation (PLOS) there are a total of 11 negative cases with 5 incorrectly assigned as positive cases, as a result the specificity (the ratio of labeled negative cases to total negative cases) is low though the overall impact to accuracy is much smaller (accuracy = 89%, F-score = 0.94). Another example occurs in code sharing (Comparator), where there are 6 true positive cases, with 5 of these correctly labeled as positive, and 5 others incorrectly labeled as positive cases (false positives). The sensitivity is relatively high (0.83), as the majority of the correct cases were labeled correctly, however the precision (and therefore the F-Score which is dependent on the precision and recall/sensitivity) is low (0.50) due to the five false positives. The overall accuracy is still very high in this open science indicator (94%), indicating only a few cases had a strong impact on the precision and F-Score metrics (precision = 0.50, F-Score = 0.64).

These accuracy metrics are excellent tools to give greater context of the strengths and weaknesses of an individual process, but need to be viewed with additional context to gauge the reliability of the metric. Due to this we prefer to provide accuracy in general which is easier to interpret and is more robust to unbalanced datasets. To give additional context to these metrics we also provide the confusion matrices that have the total of true positive, true negative, false positive, and false negative cases within each set and metric.



## PLOS Corpus:

Table 1: Proportion of articles sharing data or code (either in an online repository or as supplemental material), as a proportion of either the number of articles *generating* data or code (Manual Annotation and DataSeer only) or the total number of articles. ODDPub does not estimate whether an article generates data or code, only shares, and so is only included in the second proportion (i.e. sharing/total). These proportions are estimated with the groundtruth subset of the PLOS corpus manually annotated by DataSeer.

| <b>Sharing/Generating</b> | Manual Annotation | DataSeer      | ODDPub        |
|---------------------------|-------------------|---------------|---------------|
| Data                      | 68/86 = 79.1%     | 60/95 = 63.2% | NA            |
| Code                      | 18/41 = 43.9%     | 16/41 = 39.0% | NA            |
| <b>Sharing/Total</b>      | Manual Annotation | DataSeer      | ODDPub        |
| Data                      | 68/97 = 70.1%     | 60/97 = 61.9% | 52/97 = 53.6% |
| Code                      | 18/97 = 18.6%     | 16/97 = 16.5% | 11/97 = 11.3% |

Table 2: Accuracy metrics for the PLOS corpus. Results for DataSeer analysis and ODDPub, where applicable, are provided.

| DataSeer        | Accuracy | F-Score | Precision | Recall (Sensitivity) | Specificity |
|-----------------|----------|---------|-----------|----------------------|-------------|
| Data Generation | 89%      | 0.94    | 0.89      | 0.99                 | 0.09        |
| Data Sharing    | 89%      | 0.92    | 0.98      | 0.87                 | 0.96        |
| Code Generation | 85%      | 0.85    | 0.85      | 0.85                 | 0.84        |
| Code Sharing    | 97%      | 0.83    | 0.88      | 0.78                 | 0.97        |
| ODDPub          | Accuracy | F-Score | Precision | Recall (Sensitivity) | Specificity |
| Data Sharing    | 71%      | 0.77    | 0.90      | 0.67                 | 0.81        |
| Code Sharing    | 91%      | 0.69    | 0.91      | 0.56                 | 0.99        |

Table 3: Confusion Matrix of DataSeer and ODDPub detection results for generation and sharing of research products (either in an online repository or as supplemental material). Results are shown for the groundtruth set of the PLOS corpus manually annotated by DataSeer. ODDPub only evaluates sharing and therefore only has values for data sharing and code sharing. In code sharing totals are displayed removing articles when an annotator is unable to determine if code was used (N = 17).

|                   |     | DataSeer |    | ODDpub |    |
|-------------------|-----|----------|----|--------|----|
| Data Generation   |     | yes      | no | yes    | no |
| Manual Annotation | yes | 85       | 1  | NA     | NA |
|                   | no  | 10       | 1  | NA     | NA |
| Data Sharing      |     | yes      | no | yes    | no |
| Manual Annotation | yes | 61       | 9  | 47     | 23 |
|                   | no  | 1        | 26 | 5      | 22 |
| Code Generation   |     | yes      | no | yes    | no |
| Manual Annotation | yes | 35       | 6  | NA     | NA |
|                   | no  | 6        | 31 | NA     | NA |
| Code Sharing      |     | yes      | no | yes    | no |
| Manual Annotation | yes | 14       | 4  | 10     | 8  |
|                   | no  | 2        | 77 | 1      | 78 |

## Comparator Corpus:

Table 4: Proportion of articles sharing data or code (either in an online repository or as supplemental material), as a proportion of either the number of articles *generating* data or code (Manual Annotation and DataSeer only) or the total number of articles. ODDPub does not estimate whether an article generates data or code, only shares, and so is only included in the second proportion (i.e. sharing/total). These proportions are estimated with the groundtruth subset of the Comparator corpus manually annotated by DataSeer.

| <b>Sharing/Generating</b> | Manual Annotation | DataSeer      | ODDPub        |
|---------------------------|-------------------|---------------|---------------|
| Data                      | 45/88 = 51.1%     | 44/92 = 47.8% | NA            |
| Code                      | 6/44 = 13.6%      | 10/54 = 18.5% | NA            |
| <b>Sharing/Total</b>      | Manual Annotation | DataSeer      | ODDPub        |
| Data                      | 45/99 = 45.5%     | 44/99 = 44.4% | 15/99 = 15.2% |
| Code                      | 6/99 = 6.1%       | 10/99 = 10.1% | 6/99 = 6.1%   |

Table 5: Accuracy metrics for the Comparator corpus (~6,600 articles). Results for DataSeer analysis and ODDPub, where applicable, are provided.

| DataSeer        | Accuracy (%) | F-Score | Precision | Recall (Sensitivity) | Specificity |
|-----------------|--------------|---------|-----------|----------------------|-------------|
| Data Generation | 89%          | 0.94    | 0.92      | 0.96                 | 0.3         |
| Data Sharing    | 81%          | 0.79    | 0.80      | 0.78                 | 0.83        |
| Code Generation | 92%          | 0.94    | 0.85      | 1.0                  | 0.85        |
| Code Sharing    | 94%          | 0.63    | 0.5       | 0.83                 | 0.95        |
| ODDPub          | Accuracy (%) | F-Score | Precision | Recall (Sensitivity) | Specificity |
| Data Sharing    | 65%          | 0.43    | 0.87      | 0.29                 | 0.96        |
| Code Sharing    | 98%          | 0.83    | 0.83      | 0.83                 | 0.99        |

Table 6: Confusion Matrix of DataSeer and ODDPub detection results for generation and sharing of research products (either in an online repository or as supplemental material). Results are shown for the groundtruth set of the Comparator corpus manually annotated by DataSeer. ODDPub only evaluates sharing and therefore only has values for data sharing and code sharing.

|                   |     | DataSeer |    | ODDPub |    |
|-------------------|-----|----------|----|--------|----|
| Data Generation   |     | yes      | no | yes    | no |
| Manual Annotation | yes | 85       | 4  | NA     | NA |
|                   | no  | 7        | 3  | NA     | NA |
| Data Sharing      |     | yes      | no | yes    | no |
| Manual Annotation | yes | 35       | 10 | 13     | 32 |
|                   | no  | 9        | 45 | 2      | 52 |
| Code Generation   |     | yes      | no | yes    | no |
| Manual Annotation | yes | 44       | 0  | NA     | NA |
|                   | no  | 8        | 47 | NA     | NA |
| Code Share        |     | yes      | no | yes    | no |
| Manual Annotation | yes | 5        | 1  | 5      | 1  |
|                   | no  | 5        | 88 | 1      | 92 |

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